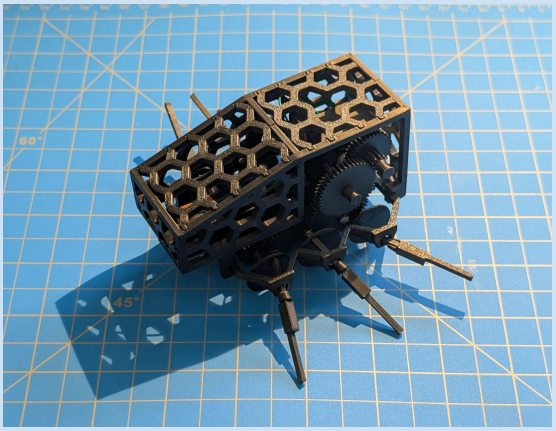
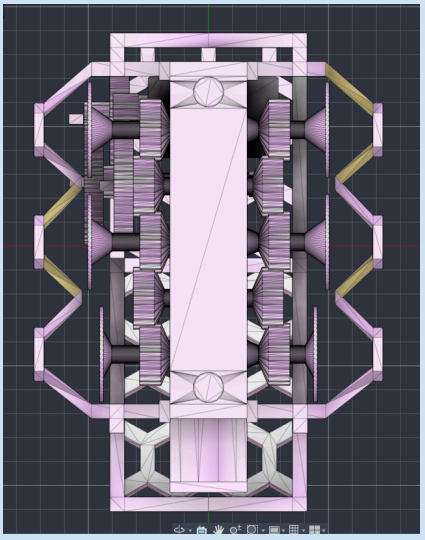
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Hexapod Rover

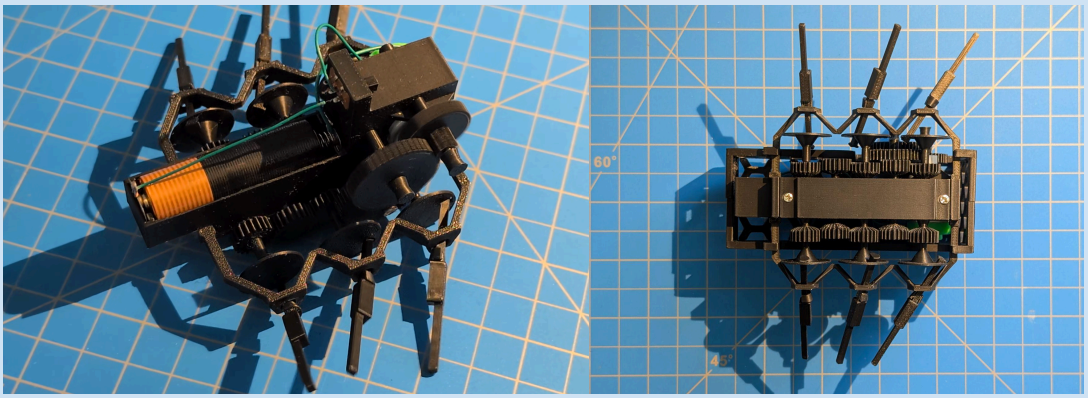


- **Fully conceptualized**, **designed**, and **constructed** a six-legged rover inspired by insect locomotion.
- Engineered a **fully custom spur gear drivetrain, exoskeleton, and linear actuators**.
- Achieved through FDM manufacturing.
- **Modular-centered design** process allows for quick repair and debugging.

Here is a short video demo on YouTube.



- The **gear system** required extensive testing and adjustments, to **find suitable tolerances**.
- The **custom gearbox** achieves a powerful **81:1** reduction ratio to **maximize torque**.
- Powered by a single **3V DC motor**.
- Achieving this amount of torque in a miniature design required careful **material and axle thickness adjustments**.

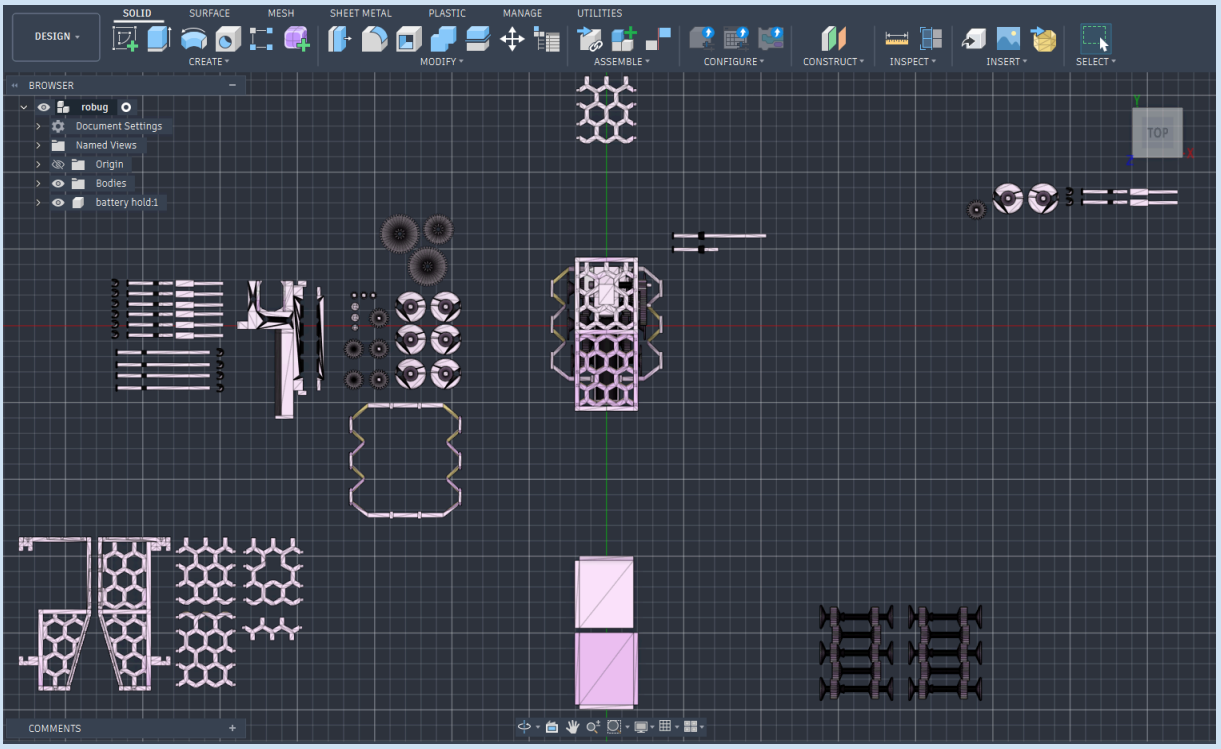


- The entire assembly is **modular**, making **troubleshooting more efficient**.
- This was especially necessary for the gear system and legs, which can be **easily disassembled** with screws.

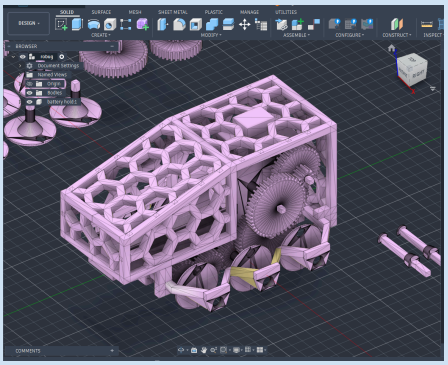
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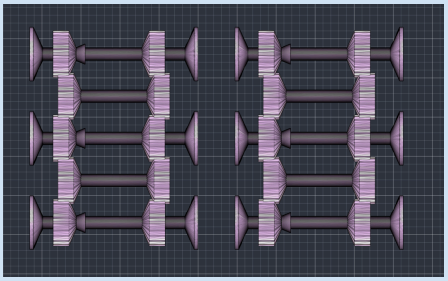
Here is a **short video demo** of the **internal mechanisms** running in the rover.



A complete top-down view of the **CAD workspace**, with disassembled parts in **proper printing orientation** displayed.



The **exoskeleton** was **optimized to save material**, while acting as a **structurally sound shell**. Instead of a solid shell, the hexagonal cutouts reduced the weight by 50%, down to 13 g, which contributed towards a lighter body in total for the legs to support. It is also the pivot for the **linear actuator legs.**



The **greatest engineering challenge** during this project was perfecting the gear system. FDM manufacturing created **huge limitations** in tolerances and part quality. Gears wouldn't fit due to **poor** print quality on **overhang surfaces**. The largest improvement arose from a unique solution: **horizontally-staggered gears.** This moved overhangs away from where gear teeth would touch, and ensured that any interlocking teeth were printed properly.

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The project went through **many design cycles** to troubleshoot errors. This gave rise to **many iterations** of the **Hexapod Rover.**

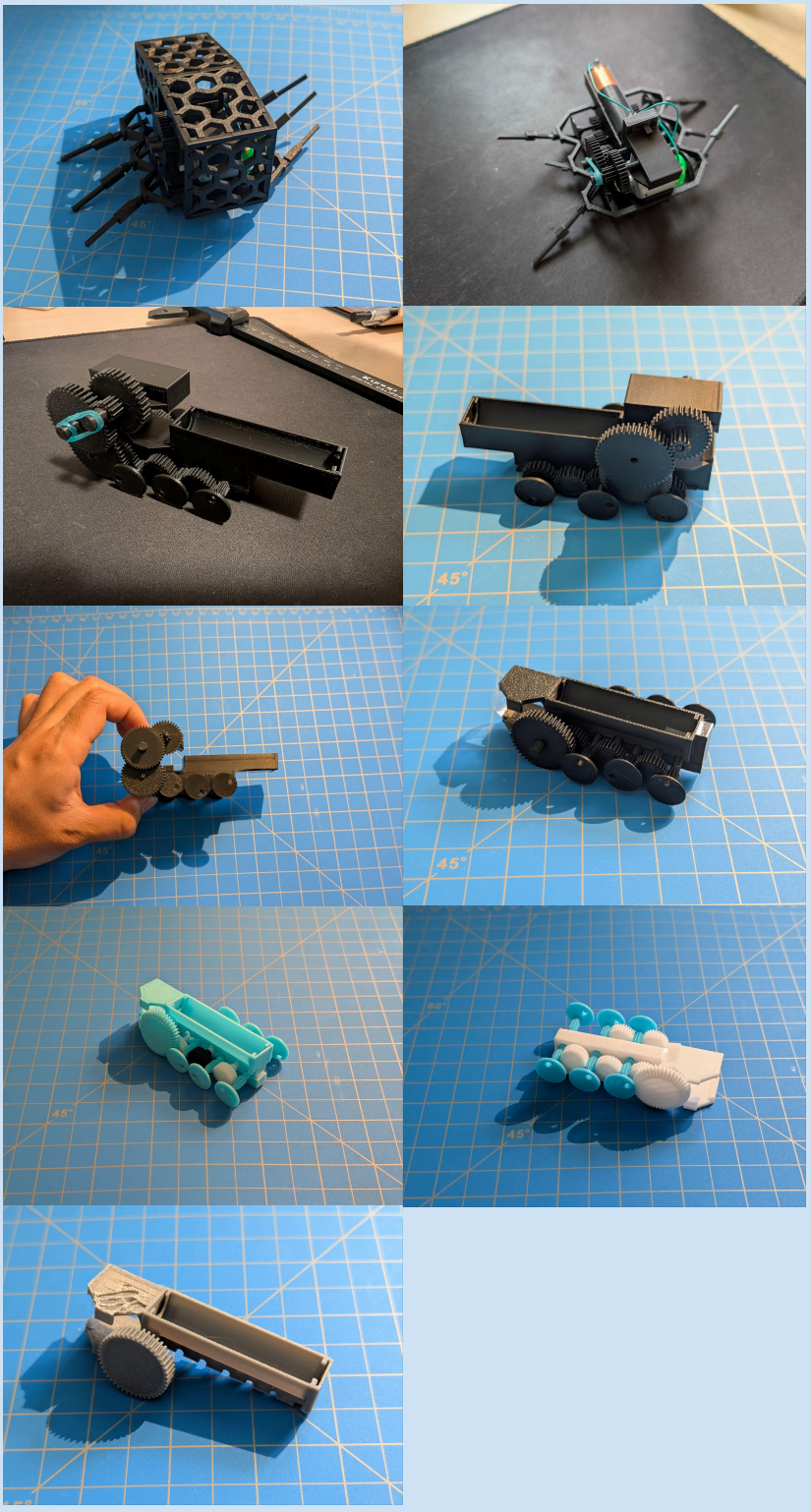
Right to left, top to bottom, images showcase regressing eras of the project's timeline.

Changes were **constantly made** to fix issues with **tolerances**, **PLA strength**, **gear teeth**, and **gear fit**.

Isolated, larger challenges included designing the **linear actuators** for the legs, **motor placement**, and the creation of the **reduction gearbox**.

Moving the motor from the back to the front required **creative gear placements** to **keep the drivetrain small** and compact.

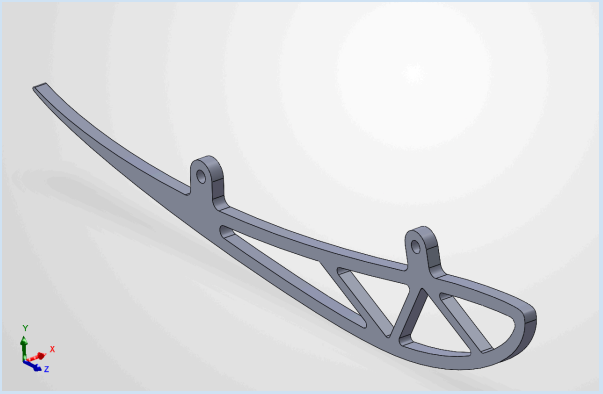
The final model of the Hexapod Rover weighed in at **95 g**, with dimensions of **134 x 52 x 102 mm**.



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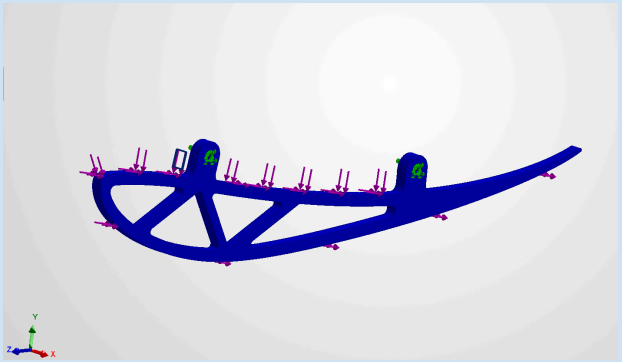
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Formula SAE Rear Wing Mount

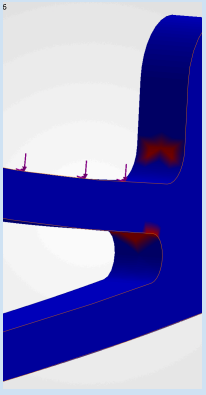


- Modelled an **aluminum mounting piece** to connect a rear spoiler wing to the frame of the car.
- Designed **specifically around the limitations** of **waterjet manufacturing**.
- Got the opportunity to learn **CAD in SolidWorks** by transferring existing skills from Fusion360.
- Learned to analyse parts under real-world stresses using **FEA** simulations.

Designing Using FEA Analysis



- By **working with the Data Acquisition** subteam, we gathered the known drag and downforces, and **ran simulations** on **various designs** of the aluminum mount.
- The simulation was **run under a safety factor of 2.57**, which allowed for **many variations** of web
- designs that sufficed under these constraints. We chose to **further differentiate by total weight** to deduce the best design.



- **A major challenge** that arose when designing for a **small form factor** like this, was that **sharp corners and edges** would always **reveal a safety factor below the 2.57 threshold**.
- **As shown in the image**, **fillets were added** to try and solve the issue of sharp corners. However, even fillets with **massive radii** still showed regions that fell below the safety factor. **Relying on fillets** meant leaving too much material on the part, **increasing its weight**.
- **Finally**, the **addition of webs** to support long and thin bridges, in **combination** with moderate radii fillets, achieved an FEA that showed zero regions that fell below the safety factor.

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